

LEARNING RESOURCE GUIDE

Generative AI for Beginners | Professional Participant Workbook

Section	Section 7: Prompt Engineering Mastery
Lecture	Lecture 7.2: Core Prompting Techniques

Learning Objectives

By the end of this lecture and its accompanying guide, you will be able to:

- Explain zero-shot, few-shot, and chain-of-thought prompting and identify when each is most effective.
- Construct few-shot prompts with appropriately selected examples for a given task.
- Apply chain-of-thought prompting to improve AI reasoning accuracy on multi-step problems.
- Use role prompting and contextual framing to shape model behaviour for professional tasks.
- Select the right prompting technique based on task complexity, consistency requirements, and output type.

Introduction

Core prompting techniques are the fundamental building blocks of prompt engineering practice. Just as a craftsman selects the right tool for each job, an AI practitioner selects the right prompting strategy based on the nature of the task: its complexity, the consistency requirements, the type of reasoning involved, and the degree of specificity needed. Knowing the techniques is necessary but not sufficient—judgment about when to apply each one separates novice from proficient prompt engineers.

The techniques covered in this lecture—zero-shot, few-shot, chain-of-thought, and role prompting—emerged from academic research into language model behaviour and have been validated extensively in production deployments. Each addresses a distinct failure mode: zero-shot fails on novel or complex formats; few-shot fails when examples are poorly chosen; chain-of-thought fails when reasoning is left implicit. Understanding these failure modes is as important as understanding the techniques themselves.

This guide provides deeper analysis of each technique, annotated examples from professional contexts, and a practical decision framework you can apply whenever you sit down with an AI tool.

Core Concepts Explained

Zero-Shot Prompting

Definition: Zero-shot prompting presents a task instruction to the model without any demonstration examples, relying entirely on the model's pre-trained knowledge of the task type.

Business relevance: Zero-shot prompting is the fastest approach and sufficient for the majority of routine professional tasks—translation, summarisation, classification of common categories, and standard writing formats. Its efficiency makes it the default starting point for any new task.

AI Practitioner relevance: AI practitioners use zero-shot prompting as the baseline and escalate to more complex techniques only when output quality is insufficient. This keeps workflows lean and prompt libraries simple.

Real-world example: Google's AI-assisted customer support tools initially relied on zero-shot prompts for standard query categorisation (billing, technical, general) because the categories were common enough that the model classified them reliably without examples.

Risks or limitations: Zero-shot prompting performs poorly on tasks requiring unusual formats, domain-specific jargon not common in training data, or novel output structures. Practitioners who default to zero-shot for complex tasks consistently receive generic or misformatted outputs.

Few-Shot Prompting

Definition: Few-shot prompting includes two to five demonstration examples in the prompt—showing the model the desired input/output pattern—before presenting the actual task. The model follows the demonstrated pattern.

Business relevance: Few-shot prompting is particularly valuable for consistency-critical tasks: classification systems where category definitions are nuanced, structured data extraction where fields must match a schema, or content generation where a specific house style must be maintained.

AI Practitioner relevance: For practitioners building repeatable AI workflows—automated document processing pipelines or content generation systems—few-shot prompts embedded in system configurations ensure consistent outputs even as individual task instances vary.

Real-world example: Klarna's AI customer service system uses few-shot prompting to classify customer intent across dozens of categories. By including curated examples of ambiguous cases (complaints that overlap billing and product quality), the system correctly classifies edge cases that zero-shot approaches routinely mishandled.

Risks or limitations: Example selection significantly affects performance. Poorly chosen examples—unrepresentative, too similar to each other, or introducing unintended patterns—can bias the model toward incorrect outputs. Practitioners should curate examples carefully and test with diverse task instances.

Chain-of-Thought (CoT) Prompting

Definition: Chain-of-thought prompting instructs the model to reason through a problem step by step before providing a final answer, either by including a worked example with visible reasoning or by appending 'Let's think step by step' to the prompt.

Business relevance: CoT prompting substantially improves accuracy on multi-step reasoning tasks: financial calculations, logical inference, legal analysis, and complex troubleshooting. For any task where intermediate steps matter—not just the final answer—CoT is the appropriate technique.

AI Practitioner relevance: AI practitioners deploying models for analytical support—risk assessment, root-cause analysis, scenario modelling—should default to CoT prompting to make the model's reasoning auditable. Visible reasoning steps make it easier to catch errors before acting on outputs.

Real-world example: McKinsey's AI research tools use chain-of-thought prompting when using language models to assess strategic options, requiring the model to articulate the logical chain from evidence to recommendation. This makes reasoning auditable during human review.

Risks or limitations: CoT prompting increases token consumption and response latency—less suitable for high-volume, low-stakes tasks. The model may also produce a confident but incorrect reasoning chain, making the error more elaborate rather than eliminating it. Human review of the reasoning chain remains essential.

Role Prompting

Definition: Role prompting assigns the model a specific identity or professional perspective at the start of the prompt ('You are a senior data scientist') to shape the register, assumed knowledge, and framing of its outputs.

Business relevance: Role prompting is one of the most practically impactful techniques because it adjusts multiple output dimensions simultaneously—vocabulary, depth, tone, and assumed audience—with a single instruction. It is particularly effective for generating professional-register outputs without extensive additional instructions.

AI Practitioner relevance: Practitioners who build AI into team workflows often embed role prompts in system-level configurations, so every interaction inherits the appropriate professional persona without requiring individual users to set it up. This creates consistent quality across a team.

Real-world example: Notion's AI writing assistant allows users to set an 'editor persona', which functions as a role prompt. When set to 'senior technical editor', the model produces more precise, structured suggestions with fewer colloquialisms—matching the quality standard of editorial teams.

Risks or limitations: Role prompts can cause the model to over-commit to a persona, occasionally generating outputs that are plausible for the role but not accurate for the actual task. A 'senior medical professional' role might produce confident medical-sounding content that is factually incorrect. Roles should be matched to the task and outputs always verified for accuracy.

Practical Applications

The following scenarios illustrate how to select and combine core techniques in professional AI workflows.

Automated Document Classification

A legal operations team classifies incoming contracts into categories (NDA, SOW, MSA, Amendment). Zero-shot prompting produces acceptable accuracy (~80%) on standard formats but misclassifies hybrid documents. Adding three curated few-shot examples per category—including hybrid type examples—raises accuracy above 95% and reduces manual review queue by two-thirds. The few-shot prompt is embedded in the processing pipeline; no user needs to construct it individually.

Financial Analysis with Auditable Reasoning

An investment analyst uses CoT prompting to analyse a company's free cash flow trend: 'Given the following five years of FCF data: [data]. Reason step by step about the trend, identify any anomalies, and state what the pattern implies for near-term capital allocation.' The model's step-by-step reasoning reveals an assumption about capex that the analyst evaluates and corrects before including the analysis in a client report.

Cross-Team Communication Adaptation

A product manager communicates the same technical update to three audiences: an engineering team, an executive committee, and a customer-facing support team. Using role prompting—'You are a product manager communicating to [audience]'—with the same base content, they generate three appropriately calibrated versions in minutes. Tone, depth, and terminology adjust automatically to each audience.

Best Practices

- **Start with zero-shot and escalate only when needed.** Zero-shot is the lowest-cost technique. Move to few-shot when you need consistency or pattern adherence, and to CoT when you need auditable multi-step reasoning.
- **Curate few-shot examples carefully.** Examples should represent the full range of inputs the model will encounter, including edge cases. Unrepresentative examples create blind spots in model performance.
- **Make CoT reasoning visible before acting on conclusions.** When using chain-of-thought, always read the reasoning chain, not just the final answer. Errors in the chain invalidate the conclusion even if the final statement sounds plausible.
- **Combine techniques when tasks are complex.** Few-shot and role prompting can be combined: assign a role and then provide examples in that role's voice. CoT can be combined with few-shot by including examples that show the reasoning process, not just the answer.
- **Test prompts with adversarial inputs.** Before deploying any prompt in a production workflow, test it with edge cases and unusual inputs. Techniques that perform well on typical cases often fail on outliers.

Common Mistakes and Misconceptions

Misconception: Using too many few-shot examples.

Correct approach: Two to five well-chosen examples typically outperform ten or more, because excessive examples consume context window space and may introduce inconsistencies. Quality of examples matters more than quantity.

Misconception: Applying CoT prompting to simple factual tasks.

Correct approach: Chain-of-thought adds cost and latency without benefit for simple lookups or straightforward tasks. Reserve it for reasoning-heavy tasks: multi-step calculations, logical inference, or structured analysis where intermediate steps are meaningful.

Misconception: Assuming a role prompt eliminates the need for accuracy verification.

Correct approach: Role prompts shape register and style, not factual accuracy. A model assigned the role of 'expert cardiologist' still cannot verify medical facts against clinical databases. Role prompts improve output quality; they do not substitute for domain expertise.

Misconception: Selecting examples that all come from the same sub-category.

Correct approach: If few-shot examples are too similar, the model learns a narrow pattern and fails on inputs that differ even slightly. Examples should span the realistic distribution of inputs the task will encounter.

Decision Framework: Technique Selection Guide

Use this table to select the most appropriate prompting technique for a given task. Start from the left column and match your task characteristics.

Task Characteristic	Recommended Technique	When to Escalate
Common, straightforward task	Zero-shot	If output is generic or off-format
Requires specific format or pattern	Few-shot (2-5 examples)	If edge cases are misclassified
Multi-step reasoning required	Chain-of-thought	If reasoning chain itself contains errors
Professional register/domain needed	Role prompting	If accuracy is critical (add verification)

Complex, multi-dimensional task	Combine: Role + Few-shot or CoT	Always verify combined technique outputs
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How to use this: Escalate to a more complex technique only when the simpler technique produces inadequate output. Unnecessary complexity increases cost and prompt maintenance burden.

Knowledge Check

Test your understanding before moving on. Answers are shown below each question.

1. A team needs to classify customer support tickets into 12 categories, including several that are frequently confused. Which technique most likely maximises classification accuracy?

- A. Zero-shot prompting with a list of categories.
- B. Few-shot prompting with curated examples of the most commonly confused categories.
- C. Chain-of-thought prompting asking the model to reason about ticket intent.
- D. Role prompting assigning the model the role of a support specialist.

Answer: B. Few-shot prompting with curated examples of the most commonly confused categories.

Why: Few-shot prompting with examples specifically targeting the confusion cases directly addresses the classification weakness that zero-shot would likely reproduce.

2. An analyst asks AI to calculate net present value and receives a single number with no explanation. The analyst cannot verify whether the calculation is correct. What should they do?

- A. Accept the output—the model is unlikely to make arithmetic errors.
- B. Re-submit the prompt adding 'Let us think step by step' to generate an auditable reasoning chain.
- C. Switch to a different model that shows its work automatically.
- D. Ask the model to simplify the calculation.

Answer: B. Re-submit the prompt adding 'Let us think step by step' to generate an auditable reasoning chain.

Why: Chain-of-thought prompting makes intermediate reasoning visible, allowing the analyst to verify each step rather than accepting an unverifiable final answer.

3. A practitioner assigns the role 'You are a world-leading oncologist' and receives a highly confident, detailed medical response. What is the most appropriate next step?

- A. Share the response directly—the oncologist role prompt ensures clinical accuracy.
- B. Verify the clinical content against authoritative medical sources before acting on or sharing it.
- C. Add more examples to the prompt to improve accuracy.
- D. Switch to chain-of-thought prompting to eliminate the hallucination risk.

Answer: B. Verify the clinical content against authoritative medical sources before acting on or sharing it.

Why: Role prompts shape register and style but do not grant the model access to verified medical knowledge. Confident-sounding clinical content must always be verified against authoritative sources.

Reflection Questions

1. Identify a professional task where you currently use zero-shot prompting. Would few-shot examples improve consistency? What examples would you include?
2. Think of a recent decision or analysis. How would you construct a chain-of-thought prompt to assist with the reasoning while keeping it auditable?
3. How could role prompting help your team communicate more effectively across different audiences—technical, managerial, or customer-facing?
4. What adversarial inputs would you use to stress-test a prompting technique before deploying it in a team workflow?
5. If you were building a shared prompt library for your team, which core technique would you document first and why?

Key Takeaways

- Zero-shot, few-shot, and chain-of-thought prompting address different task requirements; selecting the right technique is a core practitioner skill.
- Zero-shot is the fastest and lowest-cost technique; escalate only when output quality is insufficient for the task.
- Few-shot examples should be curated for representativeness—two to five well-chosen examples typically outperform a larger but poorly selected set.
- Chain-of-thought prompting makes reasoning auditable, which is essential for high-stakes analytical tasks where intermediate steps must be verified.
- Role prompting adjusts register, depth, and vocabulary simultaneously; it improves stylistic quality but does not guarantee factual accuracy.
- Combining techniques—role plus few-shot, or CoT plus role—is appropriate for complex tasks and often produces significantly better outputs than any single technique alone.
- Test prompts with edge cases before deploying them in production workflows; techniques that work well on typical inputs may fail on outliers.

Recommended Next Actions

6. Take a task you currently do with zero-shot prompting and design a few-shot version with three curated examples. Compare outputs on five different inputs.
7. The next time you use AI for a multi-step analysis, add 'Let us reason through this step by step' and read the reasoning chain before accepting the conclusion.
8. Draft a role prompt for your professional context and test it on three different task types. Document which task types benefit most from the role assignment.
9. Identify one team workflow that could be standardised with an embedded prompt using a combination of techniques. Design the prompt and test it with colleagues.